

Exploring User Expectations for AI-Mediated Second Language Communication: A Formative Participatory Design Study

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Abstract

This formative study explores how second-language (L2) speakers envision personal AI assistants (PAIAs) supporting face-to-face communication with native speakers. Through participatory sessions with 14 bilingual adults combining scenario generation, role-play, and diagramming, we surfaced preliminary tensions in AI-mediated L2 interaction: managing conversational flow, balancing transparency with naturalness, and navigating context-dependent privacy. Our exploratory findings suggest three candidate mediation patterns (solo, symmetric, and relayed assistance), each with distinct trade-offs that warrant further investigation. We contribute initial design considerations including reply-when-free defaults and role-based privacy profiles, alongside a replicable protocol for eliciting AI behavior expectations. These preliminary insights establish a foundation for future empirical work on PAIAs in multilingual communication contexts.

CCS Concepts

• **Human-centered computing** → **Scenario-based design**; *Activity centered design*; *Empirical studies in interaction design*; • **Computing methodologies** → **Intelligent agents**.

Keywords

Formative study, personal AI assistants, second language communication, participatory design, exploratory research

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1 Introduction

Personal AI assistants (PAIAs) are increasingly envisioned as companions in everyday social situations, yet their potential role in second-language (L2) face-to-face communication remains underexplored. While prior work examines AI in language education [6, 7] and multi-agent assistant scenarios [13, 14], we lack understanding

of how L2 speakers *expect* PAIAs to behave during live conversation with native speakers. This formative study addresses this gap through an exploratory investigation of a triadic configuration: an L2 speaker, a native speaker, and an AI assistant. We employed participatory design methods to surface user expectations rather than evaluate existing systems. Our approach is necessarily preliminary given the absence of deployed PAIAs with sophisticated L2 support capabilities.

We pursued three exploratory research questions. First, what expectations do L2 speakers express regarding PAIA management of interruptions and conversational flow? Second, what privacy and transparency preferences emerge when participants envision AI-mediated communication? Third, what mediation patterns do participants spontaneously generate, and what trade-offs do they perceive?

This work contributes preliminary insights into user expectations for PAIA behavior in L2 contexts, initial design considerations for future prototype development, and a replicable participatory protocol for early-stage exploration of AI-mediated communication. We explicitly acknowledge the formative nature of this research and frame our findings as hypotheses for future empirical investigation.

2 Background

2.1 AI in Language Learning

Recent work demonstrates AI's potential in language education. Ruan et al. [11] found that conversational AI improved vocabulary acquisition and speaking fluency among Chinese college students. Kim et al. [7] showed that chatbot framing (growth versus fixed mindset) influences learning outcomes. Immersive technologies have also shown promise: Bendarkawi et al. [2] found that AR-based conversation practice reduced speaking anxiety, while Zhao et al. [16] demonstrated LLM-based games could support oral communication for immigrant learners. However, these systems primarily support asynchronous practice rather than live, face-to-face communication. Authentic live communication introduces complex cognitive demands for learners, specifically concerning rapid lexical access, language switching, and self-monitoring during bilingual speech production (Baus et al. [1], Costa and Santesteban [3]). Regarding another type of AI assistant are voice user interfaces (VUIs); L2 encounter significant barriers with background noise and recognition of non-English accents Pyae and Joelsson [9]. The design of AI assistants has different requirements for L2 than native speakers Wu et al. [15]. Our study extends this work by exploring expectations for PAIAs operating during authentic interaction.



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2.2 Theoretical Orientation

Rather than advancing a complex theoretical framework, we draw selectively on established concepts to inform our interpretation. Media Richness Theory [4] suggests that face-to-face communication's high information capacity may be disrupted by AI intervention. Goffman's [5] work on face management helps explain concerns about visible AI assistance affecting perceived competence. We apply these lenses interpretively rather than testing specific hypotheses derived from them.

3 Methods

3.1 Study Design and Scope

We conducted a formative, exploratory study using participatory design methods [8, 12]. This approach is appropriate for investigating interaction paradigms where mental models are nascent. We explicitly note that our study explores user *expectations* rather than evaluating functional systems.

3.2 Participants

Fourteen bilingual adults ($n=14$) participated across two focus group sessions. Participants were recruited through convenience and snowball sampling via local community groups. Inclusion criteria required self-identification as bilingual with English as a second language. Our sample represented 10 native languages across 8 countries of origin (Spanish (Mexico and Argentina), Arabic(Egypt), Persian (Iran), Nepalese (Nepal), French(Senegal), Yoruba(Ivory Coast), (India:) Hindi, Tegulu, Kannada, Malayalam.) To contextualize language background while maintaining participant privacy regarding academic records, we gathered self-reported immersion data. Specifically, 8 out of 14 participants reported residing more than two years in an English-speaking country while affiliated with a university, 13 out of 14 reported speaking English daily, and 8 out of 14 reported writing in English daily. We acknowledge that residence duration and self-reported frequency are utilized here as indicators of immersive language exposure rather than standardized proficiency scores. This approach was chosen to surface a broad range of user expectations during this formative stage. Future research should incorporate validated proficiency assessments to enable precise analysis of how language competence shapes AI behavior expectations. The study received IRB approval, and each participant received \$25 compensation.

3.3 Protocol

Each two-hour session was conducted in English, and all participants were able to follow four sequential activities demonstrating their English proficiency.

In Activity 1 (Scenario Generation), participants documented situations where they experienced L2 communication difficulties, selecting their most compelling example for subsequent activities.

In Activity 2: Role-Play Exploration. The second activity employed role-play to explore potential PAIA behaviors using elicitation scenarios derived directly from Activity 1. These scenarios were specifically selected from the "critical incidents" documented by participants where they had previously experienced a significant

need for a translator or immediate linguistic assistance. By enacting these participant-generated situations, we aimed to observe user perspectives and surface specific expectations for how an AI assistant should behave in high-pressure contexts. To facilitate this, we defined four character personas: an L2 Speaker, a Simulated Native Speaker, a User-side PAIA, and a Relayed PAIA. The "Simulated Native Speaker" persona represented a fluent interlocutor and was performed by an L2 participant from a different linguistic background to create the necessary communicative pressure. We explicitly clarify that no actual native English speakers participated; the "Native Speaker" was a simulated persona used to test AI behavior in scenarios where participants had historically felt a need for mediation. This approach allowed us to focus on uncovering user needs rather than evaluating the linguistic accuracy of the role-play itself.

In Activity 3 (Diagramming), participants created visual representations of communication flows they enacted or imagined.

In Activity 4 (Preference Prioritization), participants identified contexts where they would most or least prefer AI assistance and tasks they would or would not delegate.

3.4 Analysis

Two researchers independently reviewed transcripts and artifacts using affinity mapping [10]. Initial coding produced 112 discrete items across 18 emergent categories. Inter-rater reliability reached substantial agreement (Cohen's $\kappa = 0.783$). Disagreements (20 items, 17.9%) were resolved through collaborative discussion. A third researcher reviewed 16% of reconciled codes. We emphasize that our analysis is interpretive and exploratory. The patterns we report represent preliminary themes warranting future investigation rather than validated findings.

4 Preliminary Findings

Our analysis surfaced four thematic areas. We present these as preliminary observations that suggest directions for future research.

4.1 Flow Management Concerns

Participants expressed strong concerns about AI interventions disrupting conversational flow. One participant described losing "the thread of conversation" after an alert, estimating 10 to 20 seconds to regain context. This sensitivity appeared across 87 of 112 coded items (77.7%).

Two potential mitigation strategies emerged from discussions. The first, which we term hold-and-release, involves queuing assistance for delivery during natural pauses. The second involves peripheral cues that provide unobtrusive indicators that support is available without demanding immediate attention. These strategies require empirical validation to assess their effectiveness and acceptability.

4.2 Context-Dependent Privacy Preferences

Participants consistently distinguished between professional and personal contexts. One participant stated: "If it is professional, I may ask AI... read this mail. If it is any personal person, I don't want to give any chance to AI."

This suggests potential value in role-based privacy profiles (such as Work, Family, or Off modes) with distinct permissions for content access and notification behavior. However, the specific configurations and their usability require systematic investigation.

4.3 Mediation Pattern Diversity

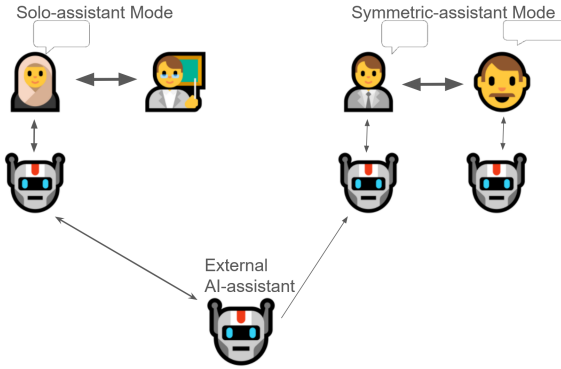


Figure 1: Schematic representation of mediation patterns explored during role-play. Left: symmetric-assistant mode (both parties have PAIAs). Right: solo-assistant mode (only L2 speaker has PAIA). Bottom: external AI communicating only with personal AIs. Icons of this image are from Miro.com

Participants' diagrams (10 completed; 4 declined due to time constraints) revealed diverse conceptualizations of AI mediation (Figure 1). These included solo assistance where AI supports only the L2 speaker, symmetric assistance where both parties have coordinating AIs, and relayed assistance where messages pass through AI intermediaries.

Participants perceived different trade-offs across these patterns: direct assistance felt faster but risked exposing private context, while relayed assistance promised better translation but obscured authorship ("Why didn't they talk directly to us?"). This diversity suggests users may lack unified mental models of AI-mediated communication, indicating a need for systems that support multiple interaction patterns.

4.4 Writing as Conversation Extension

Participants distinguished between real-time speaking assistance and post-conversation writing tasks. They expressed interest in PAIAs capturing conversational context to scaffold subsequent written communication, such as follow-up emails or documentation. This conversation-adjacent writing support emerged as a complementary need warranting further exploration.

4.5 Summary of Coded Categories

Table 1 presents the distribution of coded items across major categories, showing inter-rater agreement levels.

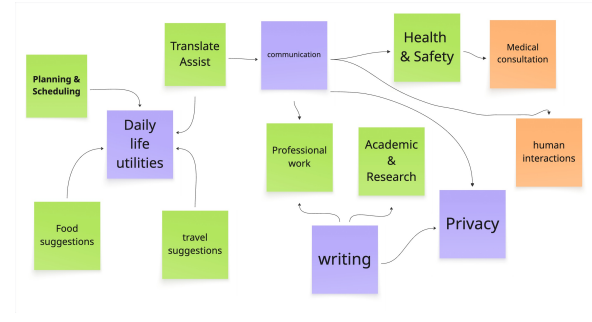


Figure 2: Affinity map showing preliminary thematic organization. Purple indicates main themes, green indicates sub-themes, and orange indicates contexts where participants preferred no AI assistance.

Table 1: Selected Categories from Affinity Analysis

Category	Items	Agreement
Communication and Quick Actions	19	78.9%
Privacy	17	94.4%
Academic	13	84.6%
Writing	9	77.8%
Planning and Scheduling	7	87.5%
Professional Work	7	77.8%
Human Interaction	7	75.0%
Medical	5	100.0%
Personal	5	80.0%
Translation and Language	4	100.0%

Note: Cohen's $\kappa = 0.783$ (substantial agreement). Only categories with four or more items shown.

5 Discussion

5.1 Preliminary Design Considerations

Based on our exploratory findings, we propose initial design considerations as hypotheses for future prototype development and evaluation.

The first consideration (DC1) concerns configurable intervention timing. Systems might benefit from reply-when-free defaults that queue assistance for natural conversational pauses, though the effectiveness of specific timing strategies requires empirical testing.

The second consideration (DC2) addresses role-based privacy profiles. Distinct permission sets for different contexts (work, personal, social) may address users' context-dependent privacy preferences. The optimal granularity and interface design remain open questions.

The third consideration (DC3) involves transparent mediation indicators. When AI transforms or routes messages, explicit attribution (such as "sent via AI") may address accountability concerns, though this could also affect conversational naturalness.

The fourth consideration (DC4) emphasizes support for multiple mediation patterns. Given the diversity of participant-generated configurations, systems may need to accommodate solo, symmetric, and hybrid assistance modes.

5.2 Limitations and Future Work

This study employed methods appropriate for formative, exploratory research aimed at generating hypotheses. Our sample of fourteen participants provided sufficient depth for qualitative saturation within our thematic analysis, consistent with established guidelines for participatory design research [12]. The diversity of native languages (10 languages across 8 countries) strengthens the transferability of our findings across L2 speaker populations, though future studies with larger samples could examine whether specific language backgrounds shape expectations differently.

Our use of self-reported language use frequency and residence duration, rather than standardized proficiency assessments, reflects an intentional methodological choice to avoid collecting protected educational records while still contextualizing participants' language backgrounds. This approach is well-suited to our exploratory aims, where the goal was to surface the range of user expectations rather than correlate expectations with precise proficiency levels. Future confirmatory research could incorporate validated assessments to examine proficiency as a moderating variable.

The role-play methodology, in which participants simulated AI behavior and enacted various roles including native speakers, represents a strength of participatory design for exploring novel interaction paradigms where functional systems do not yet exist. This approach enabled participants to externalize tacit assumptions and iteratively refine their expectations through embodied experience. While expectations may evolve when users encounter functional prototypes, our findings establish baseline user mental models that can inform initial system design. Our focus on English as the target language and recruitment from a university community provided a coherent context for examining PAIA expectations in academic and professional settings. Future research could extend these findings by examining other language pairs and community contexts to assess the breadth of applicability.

6 Conclusion

This formative study offers preliminary insights into how L2 speakers envision AI assistance during face-to-face communication. Through participatory methods, we surfaced initial themes around flow management, context-dependent privacy, and diverse mediation patterns. We contribute a replicable protocol for early-stage exploration and propose design considerations as hypotheses for future work. We explicitly frame these findings as exploratory rather than definitive. The tensions and patterns we identified (between efficiency and naturalness, transparency and discretion, assistance and autonomy) establish a foundation for future empirical investigation of PAIAs in multilingual communication contexts.

References

- [1] Cristina Baus, Mikel Santesteban, Elin Runnqvist, Kristof Strijkers, and Albert Costa. 2020. Characterizing lexicalization and self-monitoring processes in bilingual speech production. *Journal of Neurolinguistics* 56 (2020), 100934.
- [2] Jad Bendarkawi, Ashley Ponce, Sean Chidozie Mata, Aminah Aliu, Yuhuan Liu, Lei Zhang, Amna Liaqat, Varun Nagaraj Rao, and Andrés Monroy-Hernández. 2025. ConversAR: Exploring Embodied LLM-Powered Group Conversations in Augmented Reality for Second Language Learners. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–11.
- [3] Albert Costa and Mikel Santesteban. 2004. Lexical access in bilingual speech production: Evidence from language switching in highly proficient bilinguals and L2 learners. *Journal of memory and Language* 50, 4 (2004), 491–511.
- [4] Richard L. Daft and Robert H. Lengel. 1986. Organizational information requirements, media richness and structural design. *Management science* 32, 5 (1986), 554–571.
- [5] Erving Goffman. 2023. *The presentation of self in everyday life*. Routledge. 450–459 pages.
- [6] Fatih Karataş, Faramarz Yaşar Abedi, Filiz Ozek Gunyel, Derya Karadeniz, and Yasemin Kuzgun. 2024. Incorporating AI in foreign language education: An investigation into ChatGPT's effect on foreign language learners. *Education and Information Technologies* 29, 15 (2024), 19343–19366.
- [7] Minsol Kim, Aliea L. Nallbani, and Abby Rayne Stovall. 2024. Exploring LLM-based Chatbot for Language Learning and Cultivation of Growth Mindset. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–5.
- [8] Michael J. Muller. 2003. *Participatory design: The third space in HCI*. CRC press.
- [9] Aung Pyae and Tapani N. Joellsson. 2018. Investigating the usability and user experiences of voice user interface: a case of Google home smart speaker. In *Proceedings of the 20th international conference on human-computer interaction with mobile devices and services adjunct*. 127–131.
- [10] Dario Rodighiero. 2021. *Mapping Affinities*. Métis Presses. doi:10.37866/0563-99-9
- [11] Sherry Ruan, Liwei Jiang, Qian Yao Xu, Zhiyuan Liu, Glenn M. Davis, Emma Brunsell, and James A. Landay. 2021. Englishbot: An ai-powered conversational system for second language learning. In *Proceedings of the 26th International Conference on Intelligent User Interfaces*. 434–444.
- [12] Elizabeth B.-N. Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *Co-design* 4, 1 (2008), 5–18.
- [13] Austin L. Toombs, Kyle Montague, Richmond Y. Wong, Robin N. Brewer, Suchismita Naik, Paul C. Parsons, Selma Sabanovic, and Derek Whitley. 2024. Future Dialogues: Personal AI Assistants and Their Interactions with Us and Each Other. In *Companion Publication of the 2024 Conference on Computer-Supported Cooperative Work and Social Computing*. 675–678.
- [14] Jash Vora, Deepak Yadav, Ronak Jain, and Jaya Gupta. 2021. JARVIS: A PC Voice Assistant. *International Journal of Advance Study and Research Work* 4, 7 (2021), 386–393.
- [15] Yunhan Wu, Daniel Rough, Anna Bleakley, Justin Edwards, Orla Cooney, Philip R. Doyle, Leigh Clark, and Benjamin R. Cowan. 2020. See what I'm saying? Comparing intelligent personal assistant use for native and non-native language speakers. In *22nd international conference on human-computer interaction with mobile devices and services*. 1–9.
- [16] Yijun Zhao, Jiangyu Pan, Yan Dong, Tianshu Dong, Guanyun Wang, Fangtian Ying, Qihang Shen, and Jiacheng Cao. 2024. Language urban odyssey: A serious game for enhancing second language acquisition through large language models. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.